Activity 4 Pre-Lab — Analysis of OTC Pain Relievers (TLC)

Organic Lab · Introduction to Thin-Layer Chromatography

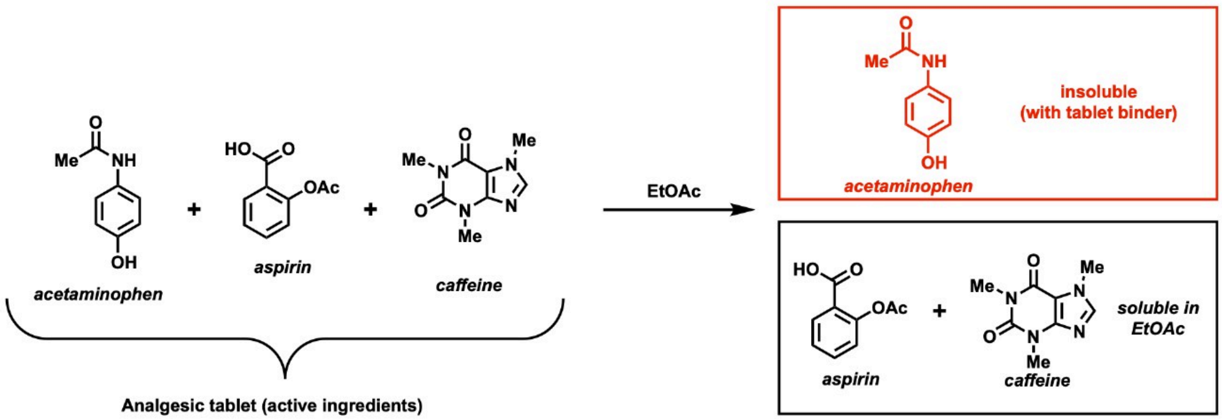
[Your name] & [partner]
[MM/DD/YY]

Black = write this in your notebook **Blue = context, don't copy it** Copy the header onto every page. Draw anything in a dashed box yourself.

Purpose

Use **thin-layer chromatography (TLC)** to separate and identify the medicinally-active compounds — **acetaminophen, aspirin (acetylsalicylic acid), and caffeine** — in an over-the-counter pain reliever (Excedrin). Compare the $\mathbf{R_f}$ values of the tablet's spots against authentic standards, and see how the **mobile-phase polarity** changes the separation.

✍ **DRAW — reproduce this scheme in your notebook (from the Activity 4 procedure)**



the three actives, and why TLC/extraction separates them: acetaminophen stays with the insoluble binder while aspirin + caffeine dissolve in EtOAc.

Chemical Reagents & Safety Table

CONTEXT ▸ Directions require a table for these 5: ethyl acetate, hexanes, acetaminophen, acetylsalicylic acid, caffeine. Values below are typical Sigma-Aldrich SDS — cross-check the SDS tile on Labflow.

Chemical	Role in lab	MW	Amount used	Physical properties	GHS pictograms
Ethyl acetate	Mobile-phase component (polar)	88.11	~20 mL (in mobile phases)	Colorless liquid; bp 77 °C; d 0.90 g/mL; flash pt −4 °C	flammable, irritant
Hexanes	Mobile-phase component (non-polar)	86.18	~16 mL (in mobile phases)	Colorless liquid; bp 69 °C; d 0.66 g/mL; flash pt −22 °C	flammable, health, irritant, environ.
Acetaminophen	Analyte / standard	151.16	trace (spotting std)	White crystalline solid; mp 168–172 °C	irritant
Acetylsalicylic acid (aspirin)	Analyte / standard	180.16	trace (spotting std)	White crystalline solid; mp ~135 °C (dec.)	irritant
Caffeine	Analyte / standard	194.19	trace (spotting std)	White solid/powder; mp 234–236 °C (sublimes ~178 °C)	irritant

FYI ▸ also used (not required in table): methanol (extract solvent — flammable/toxic) and a few drops of glacial acetic acid the instructor adds to each chamber to keep aspirin's acidic spot tight.

GRAB FIRST: mortar & pestle · 25-mL Erlenmeyer · 3× 250-mL beakers + 3 watch-glass covers · 10-mL graduated cylinder · 4 silica TLC plates · capillary spotters · pencil + ruler · forceps · UV lamp.

Procedure Plan (split the page: left = your plan, right = fill in during lab)

Plan (write before lab)

Part 1 — Find the best solvent

1. Crush **2 tablets** to powder w/ mortar & pestle → 25-mL Erlenmeyer.
2. Add **~5 mL methanol**, swirl 5 min, let the white binder settle. Clear liquid on top = your extract. Label it "T" (tablet).
3. With a graduated cylinder, make **12 mL each** of three mobile phases in labeled 250-mL beakers; cover each w/ a watch glass:
   - (i) 5:1 hexanes:EtOAc
   - (ii) 1:1 hexanes:EtOAc
   - (iii) 100% EtOAc

   Instructor adds 5 drops glacial acetic acid to each; swirl. Watch glass keeps the chamber vapor-saturated.
4. Get a 4-lane plate; **lightly** spot "T" on lane 1 of **3** plates (pencil the origin line ~1 cm up first). Too much sample streaks → unreadable. Re-spot the same dot 2–3× letting it dry between.
5. Stand one plate in each chamber (spots **above** the solvent!), re-cover. Let it run until solvent is ~1 cm from the top.
6. Remove, immediately **pencil-mark the solvent front**, air-dry.
7. View under the **UV lamp**; pencil-circle every spot. Measure distances, compute R_f, and choose the solvent with the **best-spread** spots.

Part 2 — Identify the components

8. New plate, **best solvent**, **4 lanes**: T · acetaminophen · aspirin · caffeine.
9. Develop, dry, UV-visualize, circle spots, measure each R_f.
10. Match each tablet spot's R_f to a standard → state which actives are present.

✍ **WRITE the R_f formula**

R_f = (distance spot moved from origin) ÷ (distance solvent front moved from origin)

TIP › Handle plates by the edges only; skin oils leave spots.

Experiment Notes (in lab)

Tablet brand: ____________

Part 1 — R_f per solvent:

5:1 →

1:1 →

0:1 (100% EtOAc) →

Best-resolving solvent: ____________

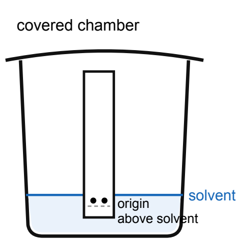
UV notes / any streaking:

Part 2 — R_f match:

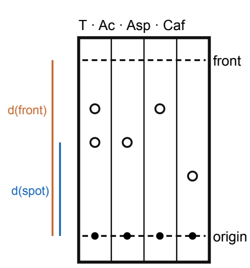
acetaminophen std:
aspirin std:
caffeine std:
tablet spots:

Actives identified in tablet:

Setup — draw these



developing chamber



developed plate + R_f

CONTEXT (don't copy) ›

Why acid? aspirin is a carboxylic acid; a little acetic acid in the mobile phase keeps it neutral so its spot stays tight.

Polarity: silica = polar stationary phase. More EtOAc (more polar) pushes everything higher (bigger R_f). Pick the solvent that spreads the three spots apart the most.

Clean-up & Conclusions

- Used solvents → **organic (non-halogenated) waste**; TLC plates, capillaries & tablet powder → solid waste; rinse & return glassware; do your assigned bench clean-up task.
- **Conclusions** (write ~4 sentences after lab): which solvent resolved the actives best & why (polarity); the R_f of each standard; which actives your tablet contained by R_f match; one error source (heavy spotting, tilted plate, running the front off).